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2021-24

Full Marks : 60

Time : 3 Hours

Candidates are required to give their answer in their own words as far as practicable. Their figures in the margin indicate full marks.

Answer from **both** the Sections as directed.

Section – A

1. Choose the **correct** answer of the following : 1×10=10

(i) The minimum number of states required to recognize an octal number divisible by 3 are/is :

(a) 1 (b) 3

(c) 5 (d) 7

(ii) A DFA cannot be represented in the following format :

(a) Transition graph

(b) Transition Table

(c) C code

(d) None of these

(iii) A finite automata recognizes :

(a) Any language

(b) Context sensitive language

(c) Context free language

(d) Regular language

(iv) The entity which generate language is termed as :

(a) Automata

(b) Tokens

(c) Grammar

(d) Data

(v) A turing machine that is able to stimulate other turing machines is :

(a) Nested turing machine

(b) Universal Turing Machine

(c) Counter Machine

(d) None of these

(vi) Concatenation operation refers to which of the following set operations :

(a) Union

(b) Dot

(c) Kleene

(d) Two of the options are correct

(vii) The value of n if turing machine is defined using n-tuples.

(a) 6 (b) 7

(c) 8 (d) 5

(viii) RASP stands for :

(a) Random Access Storage Program

(b) Random Access Stored Program

(c) Randomly Accessed Stored Program

(d) Random Access Storage Programming

(ix) While applying pumping lemma over a language we consider a string W that belong to L and fragment it into ____ parts.

(a) 2 (b) 5

(c) 3 (d) 6

(x) Maximum number of states of a DFA converted from an NFA with n states is :

(a) n (b) n^2

(c) 2n (d) None of these

2. What are Formal languages ? Explain briefly. 5

Section - B

Answer any three questions of the following : 15×3=45

3. (a) Differentiate between DFA and NFA

(b) Construct a DFA to accept strings of O's and I's ending with 101

4. What do you mean by Turing Machine ? What are the primary objectives of Turing machine ? Explain halting problem of Turing Machine with suitable example.

5. What are ambiguities in grammars and languages ? Explain briefly and also explain terminal and non-terminal symbols of a grammar.

6. (a) How NFA to DFA conversion is done? Explain with example.

(b) Explain closure properties of Regular languages.

7. What do you mean by pushdown Automata ? Explain, in detail with example.

8. Write short notes on any **three** of the following : 5×3=15

(i) Parse tree

(ii) Kleene star

(iii) Transitior graph

(iv) Pumping Lemma

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